L1.3

L1.3 Linux Mint on an External SSD

Level: Foundation (Free) **Time:** 45-90 minutes **Type:** Practical full installation **Author:** Adi Ciobanu LeYa Academy **You need:** External SSD + 8GB+ USB stick + laptop (verified in L1.2)

What you'll know by the end

- How to create a bootable USB stick with Linux Mint
- How to install Linux Mint on an external SSD without touching Windows
- How to perform first boot, system update, and install core tools

---

Quick glossary before you start

Term	What it means
ISO	An exact copy of an operating system, in a single file. Like a ZIP archive, but for entire systems.
Checksum	A digital fingerprint of a file. You verify the file wasn't corrupted during download.
Bootable	A USB stick your laptop can start from. Like a startup key for Linux.
BIOS / UEFI	The hidden menu of your laptop, accessible before Windows starts. From here you control which drive it boots from.
Secure Boot	A security feature that blocks Linux from starting. Must be disabled.
EFI	A small partition (500MB) where the laptop's bootloader lives. Created automatically during install.
ext4	The standard Linux file format. Don't overthink it it works.
GRUB	The menu that appears at startup asking: Windows or Linux?

Content

Step 1: Download Linux Mint 22.3 ISO

The file is ~2.8GB. Make sure you have a stable internet connection.

1. Go to https://linuxmint.com/download.php
2. Download the **Cinnamon** edition (most popular, familiar interface)
3. After downloading, **mandatory**: verify the checksum. It's the only way to know the file isn't corrupted.

How to verify the checksum:

```
Windows: certutil -hashfile linuxmint-22.3.iso SHA256
Linux: sha256sum linuxmint-22.3.iso
Mac: shasum -a 256 linuxmint-22.3.iso
```

Compare the result with the one on the official website (in the "Verify your ISO" section). If they don't match corrupted file delete and re-download.

Official link slow? Search "linux mint mirrors [your country]" for faster servers.

Step 2: Create a bootable USB stick

Pick the section for your operating system:

On Windows **Recommended: Rufus.** One tool, well-tested, works on any Windows version.

1. Download Rufus from https://rufus.ie

- **Windows 10/11:** latest version
- **Windows 7/8:** use Rufus 3.22 (link in the "Other versions" section on the site)

2. Plug in your USB stick
3. Open Rufus, select the downloaded ISO, click **START**
4. Wait ~5 minutes until it says "READY"

Common issues:

Issue	Solution
"Access denied"	Run Rufus as Administrator (right-click Run as Administrator). Temporarily disable antivirus.
Rufus doesn't see the USB stick	Unplug it and plug it back in. Try a different USB port. Check Windows Disk Management. If it still doesn't show defective stick.
"ISO image extraction failure"	Checksum mismatch? Re-download the ISO.

On Linux You'll use the `dd` command.

> **DANGER: `dd` has no undo.** If you write to the wrong drive, you lose EVERYTHING. Zero recovery. Read carefully.

1. Identify your USB stick: `lsblk` see all drives. The stick is usually `/dev/sdb` or `/dev/sdc`, ~8-16GB in size.
2. **CHECK THE DEVICE TWICE.** `/dev/sda` is your internal drive. Do NOT write to it.
3. Run: `sudo dd if=linuxmint-22.3.iso of=/dev/sdX bs=4M status=progress`
4. Safer alternative: **USB Image Writer** (pre-installed on Linux Mint, graphical interface).

On Mac

1. Use **Balena Etcher** (https://etcher.io) graphical interface, impossible to pick the wrong drive. Recommended.
2. Terminal alternative:

- `diskutil list` identify your stick (`/dev/diskX`)
- `sudo dd if=linuxmint-22.3.iso of=/dev/rdiskX bs=4m`
- After: `diskutil eject /dev/diskX`

Step 3: Boot preparation BIOS and Secure Boot

3.1 Disable Fast Boot in Windows (before anything else!)

1. Control Panel Power Options
2. "Choose what the power buttons do"
3. "Change settings that are currently unavailable"
4. **UNCHECK** "Turn on fast startup"
5. Save changes

Without this, your laptop completely ignores the USB stick and boots straight into Windows. This is the #1 cause of "it doesn't work."

3.2 How to enter BIOS Restart your laptop. As soon as you see the logo, start pressing the key. If Windows starts you missed it. Restart and try again.

BIOS keys vary by manufacturer:

Manufacturer	BIOS key	Boot Menu key (direct)
Dell	F2	F12
HP	F10	Esc / F9
Lenovo	F2 / Fn+F2	F12 / Enter then F12
Asus	F2 / Del	Esc / F8
Acer	F2	F12
Toshiba	F2	F12
MSI / Gigabyte	Del	F11 / F12
Samsung	F2	F10 / Esc
Sony VAIO	F2 / Assist	F11
Microsoft Surface	Hold Volume+ at startup	

Lenovo: some models have a small button ("Novo") on the side hold it for 3 seconds.

3.3 Disable Secure Boot (MANDATORY) Secure Boot blocks Linux from starting on most modern laptops. It's the #1 cause of "black screen after boot."

1. In BIOS, look for the **"Security"** or **"Boot"** tab
2. Find **"Secure Boot"** set to **Disabled**
3. On some Dell/Lenovo laptops, the option is hidden under "Advanced Boot Options" or requires a temporary BIOS password. If you can't find it Google "[laptop model] disable secure boot".

3.4 Set boot order

1. Look for "Boot Order" or "Boot Priority"
2. Move **USB to the top** (+/- or F5/F6 keys)
3. Save and exit (usually F10)

Step 4: Install on the external SSD step by step

4.1 Boot from USB

- If you see the Linux Mint menu you made it.
- Choose **"Start Linux Mint"** (not "Install" directly test first)

4.2 Check before installing In the live session: - Does WiFi work? Touchpad? Does the screen look normal?

Live session issues:

Issue	Solution
WiFi not working	Connect via Ethernet cable (temporary) or use USB tethering from your phone. Drivers install later, from Driver Manager.
Black screen / wrong resolution	Reboot. At the GRUB menu press `e`, add `nomodeset` to the line with `linux`, then Enter. After install Driver Manager.
Laptop boots straight to Windows, ignores USB	You didn't disable Fast Boot (Step 3.1) or Secure Boot is still on (Step 3.3). Go back.

4.3 The actual installation

1. Click the **"Install Linux Mint"** icon on the desktop
2. Choose language Continue
3. ** AT THE PARTITIONING STEP THE MOST IMPORTANT PART OF THE WHOLE COURSE:**

Choose **"Erase disk and install Linux Mint"** NOT "Something else".

The installer will ask **WHICH DISK**. Here's the critical moment:

- Internal drive is usually `/dev/sda` (Windows, 256GB+, recognize it by size)
- External SSD is usually `/dev/sdb` or `/dev/sdc` the size you bought
- **Check the size.** 256GB external SSD? Select the ~256GB drive.
- **IF YOU'RE NOT SURE WHICH IS THE EXTERNAL SSD STOP.** Better to ask than to wipe Windows.

4. Continue: username, password, locale, keyboard. Pick something easy to remember.

4.4 What to do if installation freezes

Situation	Action
Freezes at a percentage	Wait 5 minutes. Sometimes it looks frozen but it's working.
Nothing for 10+ minutes	Reboot, remove USB stick, reinsert it, restart the install from scratch.
Freezes again at the same spot	Corrupted ISO. Re-download and re-verify the checksum.

Step 5: First boot from SSD

1. The installer says "Remove installation media and press Enter"
2. **Remove the USB stick** and press Enter
3. You should see the GRUB menu Linux Mint starts

First boot issues: ** Boots straight to Windows (no GRUB menu):**

GRUB was installed to the wrong EFI partition. You need to reinstall it:

1. Reboot from the USB stick enter "Try Linux Mint" again
2. Open a terminal and run:

```
lsblk # identify external SSD (e.g., /dev/sdb)
sudo mount /dev/sdb2 /mnt # main partition
sudo mount /dev/sdb1 /mnt/boot/efi # EFI partition
sudo grub-install --boot-directory=/mnt/boot --efi-directory=/mnt/boot/efi /dev/sdb
sudo update-grub
```

3. Reboot without the USB stick. You should see GRUB.

If it still doesn't work: don't keep trying alone. This is a complex EFI issue. Contact support we need your exact laptop model.

Black screen after boot:

Same video driver issue as Step 4.2. Reboot, at GRUB press `e`, add `nomodeset`. After boot Driver Manager install recommended drivers.

WiFi still not working:

1. Connect via Ethernet for 5 minutes
2. Open Driver Manager from the Linux Mint menu
3. Install the recommended WiFi driver
4. No cable? USB tethering from your phone works

System update (mandatory)

```
sudo apt update && sudo apt upgrade -y
```

> The first update is the longest it can take 10-30 minutes depending on your internet speed. That's normal. Don't shut down the laptop during the update.

Step 6: Install core tools

```
sudo apt install -y git curl python3 python3-pip nodejs npm cmake build-essential
```

** Node.js from apt is often OUTDATED.** After the base install, upgrade it:

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
```

Verify: `git --version`, `python3 --version`, `node --version` (must be 18.x)

Installation issues:

Error	Solution
"Unable to locate package"	Run `sudo apt update` first
cmake error	Install manually: `sudo apt install -y build-essential cmake`
Node.js still old after upgrade	Restart terminal and re-check

---

Exercise

Install Linux Mint on your external SSD following the steps above. Rules:

- Don't skip any step each one exists because someone had a problem there
- If you get an error, **write down the exact message** it's essential for debugging
- Don't move on to L1.4 until you have: Linux booting from SSD + update done + tools installed

---

Checkpoint don't move on until you can answer

1. Did you successfully boot Linux from the external SSD without the USB stick?

Check: Restart your laptop without the USB stick plugged in. Do you see the GRUB menu? Does Linux start? If yes

2. Did you run `sudo apt update && sudo apt upgrade`?

Check: Run the command again. If it says "0 upgraded, 0 newly installed"

3. Which core tools are installed?

Check: `git --version`, `python3 --version`, `node --version`. All three return a version number?

> **If you answered NO to question 1:** don't move to L1.4. Re-read Step 5 (First boot) and the GRUB reinstall procedure. If it still doesn't boot after two attempts support. We need: laptop model, what you tried, exact error message.
>
> **If you answered NO to question 2:** don't continue. An un-updated system = security and compatibility issues.
>
> **If tools are missing:** go back to Step 6 and run the commands. If you get errors, check the issues table.

All 3 checked? Move on to L1.4 Hermes Agent: Installation.

LeYa Academy Level 1 Lesson 3 v1.0 July 2026